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Math-1730-04 Winter 2025 Solutions To TEST

1. (10 points) Find $\int \frac{1}{x(\ln x)^8} dx$. Show your work and fill the blank line with your answer.

Your answer: $-\frac{1}{7(\ln x)^7} + C$

Note. This question is similar to Example 9(7) in Section 5.5 from lecture outlines, and it is also similar to Question #25 from this section in Homework 2.

Solution. Note that $(\ln x)' = \frac{1}{x}$.

$$\text{By the GPR, } \int \frac{1}{x(\ln x)^8} dx = \int (\ln x)^{-8} \frac{1}{x} dx = -\frac{1}{7}(\ln x)^{-7} + C = -\frac{1}{7(\ln x)^7} + C.$$

If one uses the substitution $u = \ln x$, then $du = \frac{1}{x} dx$ and

$$\int \frac{1}{x(\ln x)^8} dx = \int \frac{1}{u^8} du = -\frac{1}{7}u^{-7} + C = -\frac{1}{7(\ln x)^7} + C.$$

2. (12 points) Find $\int \tan^5 \theta d\theta$. Show your work and fill the blank line with your answer.

Your answer: $\frac{1}{4} \sec^4 \theta - \sec^2 \theta + \ln |\sec \theta| + C$ or $\frac{1}{4} \sec^4 \theta - \tan^2 \theta + \ln |\sec \theta| + C$

Note. This question is Question #31 from Section 7.2 in Homework 3, and it is similar to Example 4 in this section from lecture outlines.

Solution. See the solution in Student Solutions Manual.

3. (10 points) Find $\int \frac{\tan x \sec x}{1 + \sec^2 x} dx$. Show your work and fill the blank line with your answer.

Your answer: $\tan^{-1}(\sec x) + C$

Note. This question is similar to Example 8(iv) in Section 5.5 in lecture outlines.

Solution. Let $u = \sec x$. Then $du = \tan x \sec x dx$ and

$$\int \frac{\tan x \sec x}{1 + \sec^2 x} dx = \int \frac{1}{1 + u^2} du = \tan^{-1}(u) + C = \tan^{-1}(\sec x) + C.$$

4. (12 points) Find $\int \frac{1}{x^2\sqrt{x^2+1}} dx$. Show your work and fill the blank line with your answer.

Please note that your answer should not contain a trigonometric function.

Your answer: $\underline{-\frac{\sqrt{x^2+1}}{x} + C}$

Note. This question is similar to Example 11 in Section 7.3 from lecture outlines.

Solution. Let $x = \tan \theta$ ($-\frac{\pi}{2} < \theta < \frac{\pi}{2}$). Then $\sqrt{x^2+1} = \sec \theta$, $dx = \sec^2 \theta d\theta$ and

$$\int \frac{1}{x^2\sqrt{x^2+1}} dx = \int \frac{1}{\tan^2 \theta \sec \theta} \sec^2 \theta d\theta = \int (\sin \theta)^{-2} \cos \theta d\theta = -\frac{1}{\sin \theta} + C.$$

The last equality above follows from the GPR or the substitution $u = \sin \theta$.

Now $\sin \theta = \frac{x}{\sqrt{x^2+1}}$. Therefore, we have

$$\int \frac{1}{x^2\sqrt{x^2+1}} dx = -\frac{\sqrt{x^2+1}}{x} + C.$$

5. (12 points) Find $\int \frac{1}{\sqrt{x^2-1}} dx$. Show your work and fill the blank line with your answer.

Please note that your answer should not contain a trigonometric function.

Your answer: $\underline{\ln |x + \sqrt{x^2-1}| + C}$

Note. This question is similar to Example 12 in Section 7.3 from lecture outlines.

Solution. Let $x = \sec \theta$ ($0 \leq \theta < \frac{\pi}{2}$). Then $\sqrt{x^2-1} = \tan \theta$, $dx = \sec \theta \tan \theta d\theta$, and

$$\int \frac{1}{\sqrt{x^2-1}} dx = \int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + C.$$

Now $\sec \theta = x$ and $\tan \theta = \sqrt{x^2-1}$. Therefore,

$$\int \frac{1}{\sqrt{x^2-1}} dx = \ln |\sec \theta + \tan \theta| + C = \ln |x + \sqrt{x^2-1}| + C.$$

6. (14 points) (i) Find $\int x e^{-x} dx$. Show your work and fill the blank line with your answer.

Your answer: $-x e^{-x} - e^{-x} + C$

(ii) Use (i) above to find the improper integral $\int_0^{\infty} x e^{-x} dx$. Show your work and fill the blank line with the exact answer. Your answer: 1

Note. Part (i) is similar to Example 1(5) in Section 7.1 from lecture outlines, and it is also similar to Question #5 in this section from Homework 3. Part (ii) is similar to Question #27 in Section 7.8 from Homework 4.

Solution. (i) Using IBP, we have

$$\int x e^{-x} dx = -x e^{-x} + \int e^{-x} dx = -x e^{-x} - e^{-x} + C.$$

(ii) For $t > 0$, by (i) above and FTC-Part 2, we have

$$\int_0^t x e^{-x} dx = \left[-x e^{-x} - e^{-x} \right]_0^t = 1 - \frac{t+1}{e^t}.$$

By the definition of the improper integral and using L'Hospital's Rule, we have

$$\int_0^{\infty} x e^{-x} dx = \lim_{t \rightarrow \infty} \int_0^t x e^{-x} dx = 1 - \lim_{t \rightarrow \infty} \frac{t+1}{e^t} = 1 - \lim_{t \rightarrow \infty} \frac{1}{e^t} = 1.$$